$$\mathbf{N}(\mathbf{a}^+|n\rangle) = \mathbf{a}^+(\mathbf{N}+1)|n\rangle.$$

The last step is to use the fact that |*n*> is an eigenvector of **N** with eigenvalue *n*. That means we can replace (**N** +1) with (*n* + 1):

$$\mathbf{N}(\mathbf{a}^+|n\rangle) = (n+1)(\mathbf{a}^+|n\rangle). \qquad (10.26)$$

As always, when we run on autopilot, we have to keep our eyes open for interesting results. Eq. 10.26 is interesting. It says that the vector $\mathbf{a}^+|n\rangle$ is a new eigenvector of **N** with eigenvalue (n +1). In other words, given the eigenvector |*n*>, we have discovered another eigenvector whose eigenvalue is increased by 1. All of this can be summarized by the equation

$$\mathbf{a}^+|n\rangle = |n+1\rangle. \qquad (10.27)$$

Obviously, we can do this again and again to find the eigenvectors |*n* +2>, |*n* +3>, and so on. Remarkably, we find that if there is an eigenvalue *n*, there must be an infinite sequence of eigenvalues above it, spaced by integers. The name *raising* operator seems well chosen.

What about the lowering operator? Not surprisingly, we find that $\mathbf{a}^-|n\rangle$ produces an eigenvector whose eigenvalue is one unit lower:

$$\mathbf{a}^-|n\rangle = |n-1\rangle. \qquad (10.28)$$

This suggests that there must be an unending sequence of eigenvalues below *n*, but that can't be correct. We already know that the ground state has positive energy, and because **H** = $\omega\hbar$(**N**+1/2) the downward sequence must end. But the only possible way it can end is for there to be an eigenvector |0> such that when $\mathbf{a}^-$ acts on it, the result is zero. (We should not confuse |0> with the zero vector.[3]) Symbolically, this can be expressed as

$$\mathbf{a}^-|0\rangle = 0. \qquad (10.29)$$